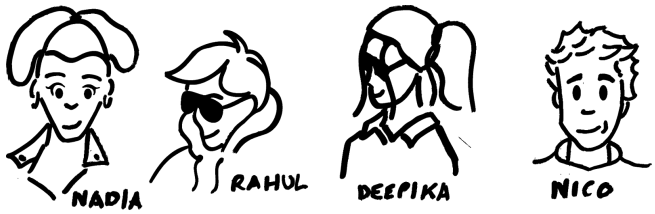
Coursework 1

ECS661U/ ECS661P User Experience Design 2024-25
Queen Mary University of London

GROUP X



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Image citation: (Fast., 2025)

Abstract

This report will cover how we conducted an ethnographic study and produced a design solution to improve user experience in gyms. We have made observations and refined our research question using grounded theory which led to defining our problem. From this we were able to make an appropriate multimodal design that takes into account the user and environment. We lastly conducted a critical comparison to an ethnographic study and design formulated by AI.

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Part 1: Ethnographic study

1.1 Methodology:

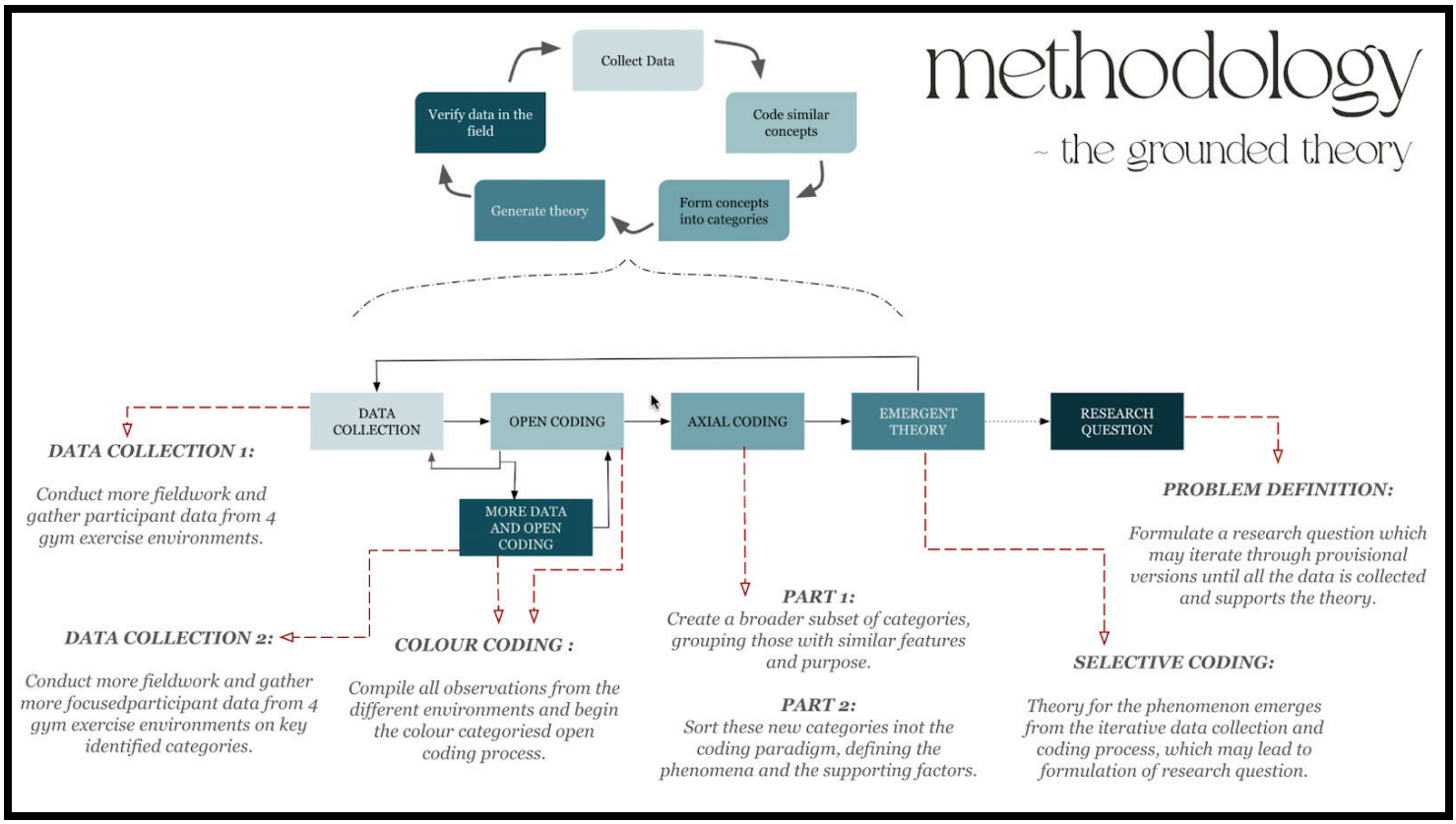
Health, fitness, and confidence. These are the key benefits of engaging in sports and exercise. Just as students gain knowledge from lectures, working out provides both immediate and long-term advantages. However, our team discovered that fitness isn't just about physical health; it also influences self-perception, efficiency, and mental well-being. Each of us had personal experiences with different fitness environments: Qmotion, The Gym Group, outdoor gym spaces, and the Arch Climbing Wall. Why does exercising in front of others impact self-esteem? How can a carefully planned routine take longer than expected? By exploring these questions, we aim to understand how different workout environments shape our experiences and what can be done to improve confidence and efficiency in fitness.

Ethnography lies within the third paradigm of human computer interaction. It is not just about the interface of the system and technologies involved in the environment but also what happens around them. This paradigm involves in depth analysis of the contextual environment and building the design to accommodate it. This phenomenological matrix characterises concerns multidimensionally, allowing for the integration of multiple theories and perspectives without exclusivity. *(Harrison, Tatar and Sengers, 2007)*

Conducting an ethnographic study on the context of gyms will allow us to execute a qualitative enquiry via first hand observations. This involves conducting field work by observing participant behavior consistently over a period of weeks across different gym environments. This process will help develop an analytic understanding on thematic interactions and behaviours drawn from the several observations made. Participant behaviour isn't always a product of their physical environment, as personal lives and context that may influence specific behaviours remain unknown. This is why we chose to observe a broader range of physical activity environments at different times of the day and week to gather authentic and raw data with a variety of demographics and functionality, to avoid bias and forced assumptions. This allows us to work with data where we can differentiate coincidental and outlier observations against those that form a pattern of behaviour. *(Hammersley, 2006)*

OUR ETHNOGRAPHIC METHODOLOGY:

A gym accumulates diverse participants with varying experience levels and behaviors. Collecting qualitative data will allow us to uncover contextual issues and interactions that lead to our defined phenomenon of behaviour in this particular environment.



Using the Grounded Theory approach of analysis will enable us to draw coherent concepts between our observations that allow us to frame and identify causal conditions to our phenomenon well. The process is inductive and iterative, where qualitative data is collected, coded, analysed, categorised and then the process repeats with refined constraints on areas to observe and recollect data. This new data is then re-coded, analysed and sub-categorised and is constantly compared to the initial provisional concepts to allow for progressive refinement of data and theory. *(Khambete and Athavankar, 2010)*

The advantage of using grounded theory is that it allows us to develop systematic theories of the phenomena and keep a record of the progress of the analysis that can be easily traced back. *(Lawrence and Tar, 2013)*

1.2 Observations:

We chose to explore various workout spaces because fitness is a personal journey, and different environments can significantly impact motivation, confidence, and effectiveness. Some of us preferred “structured” gym settings while others enjoyed the freedom of open and personal challenges. By comparing these experiences, we wanted to understand the psychological and practical differences between them and how they shape interactions.

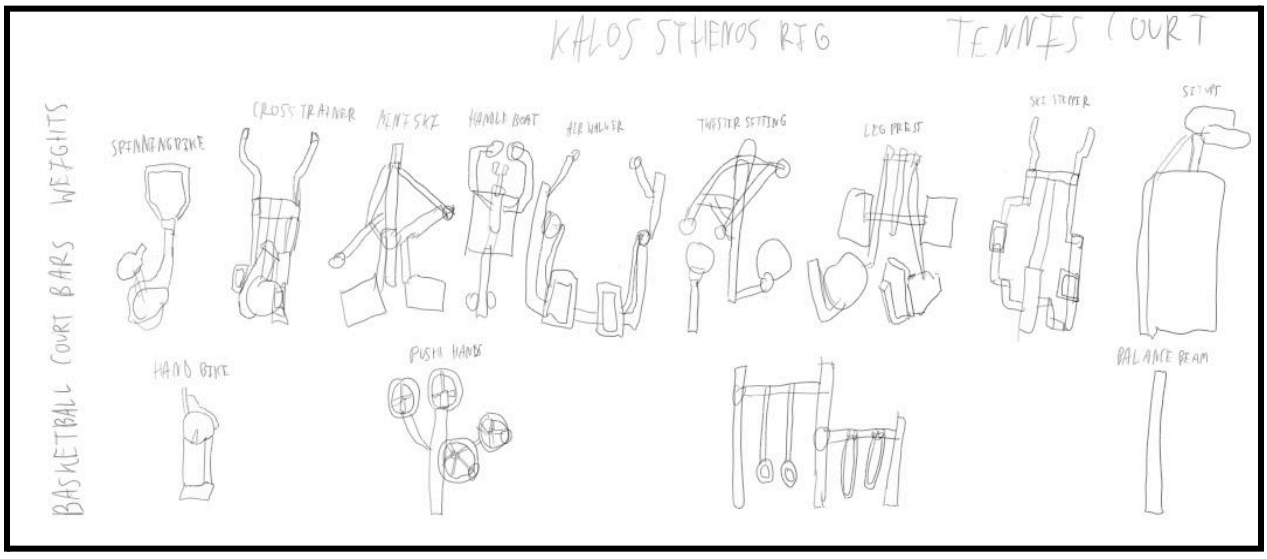
<u>Environment 1</u>: Outdoor gyms:

Outdoor public gyms consist of exercise equipment, weight machines and calisthenic bars. They are meant for youth and adults of a height bigger than 1.4 meters. Looking at demographics, elderly people were seen to just be using the main equipment area. Aside from using the machines they also stretched more than other age groups.





People also brought their own equipment such as mats and dumbbells. They were doing their own exercises with headphones on and when one person was left alone, they started playing a podcast out loud. Looking at gender, two boys saw the calisthenics bars and started competing with each other seeing who can do the most pull ups. Girls had less of a tendency to do that as two girls tried the weights machines but not for long. Outdoor gyms are overall a friendlier, less intimidating environment.

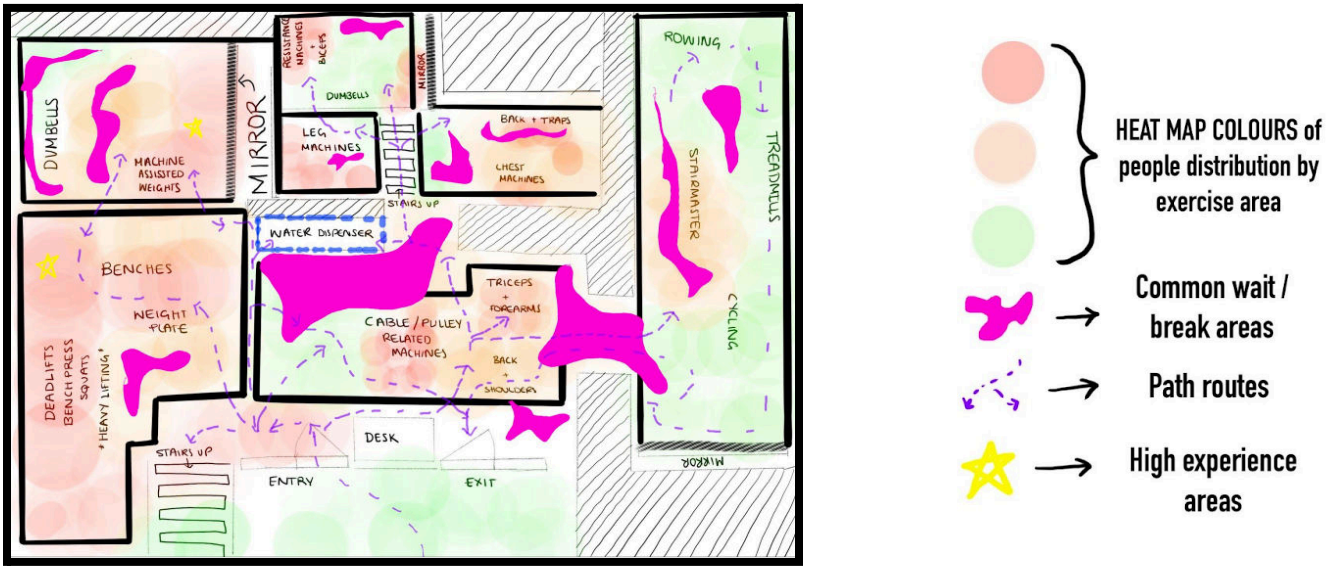


<u>Environment 2</u>: Q Motion

Observing the university gym gave an insight to the attitude of students and how their behaviours adapt in skill-hierarchy environments. The provision of the female gym provides a quiet escape from the noisier and intimidating main gym which has essentially become the “male” gym. This builds an awkward gender segregation, creating a more unapproachable environment to work in as a beginner. Conversation happens mostly amongst groups who come together, otherwise people isolate themselves with their personal devices between sets and exercises.

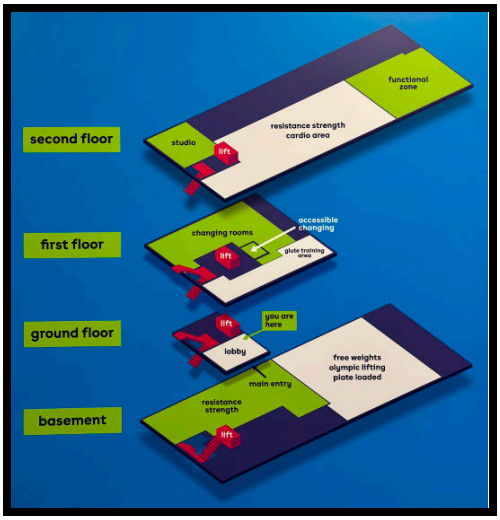
Individual participants looked much more anxious and experimented with their exercises a lot less compared to those who worked out in pairs or groups. You often find many people awkwardly walking around to see which machine or exercise is next available instead of just asking the person using it. People often mark their territory on machines or equipment by placing their belongings which makes it unusable for others between sets. A rough floor plan visual of this gym has been drawn through observing several patterns and consistent behaviours.

QMOTION MAIN GYM FLOOR PLAN:



<u>Environment 3</u>: The Gym group

Gym group is a large gym chain in the United Kingdom. We observed that the gym was really busy on weekdays especially after 5 Pm, and sometimes it was really difficult for people to find the desired machine/equipment free, causing delays and long waiting times for the people. Additionally, sometimes some equipment is difficult to find in the gym (for example, dumbbells, plates, barbell spring collar), as they are scattered all over the place.

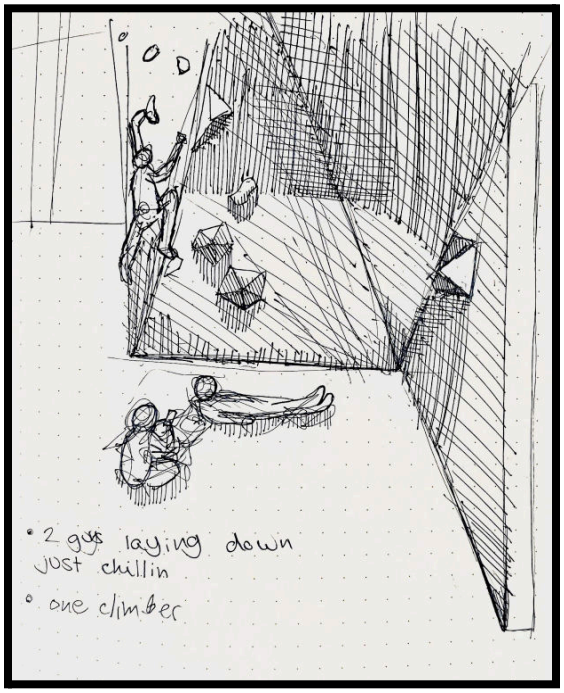


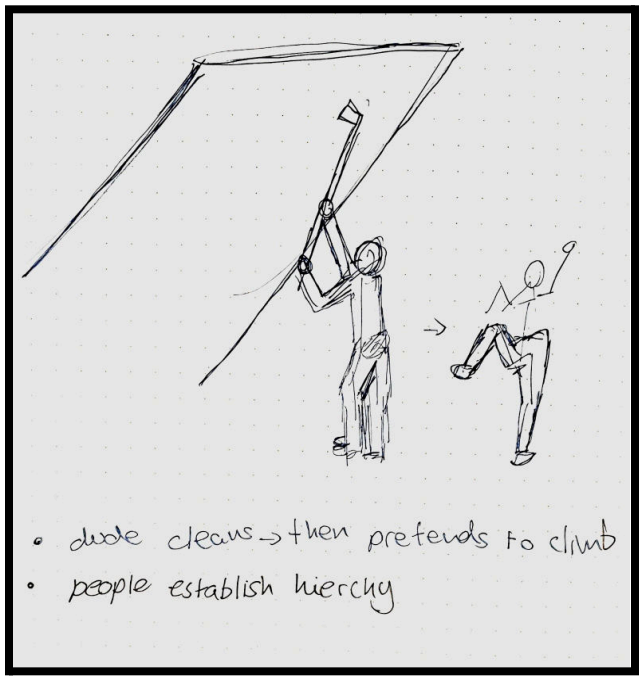
Another key observation was that we noticed there was a bit of uncertainty and anxiety amongst beginners, they were more aware of people walking by and not certain on which or how to do an exercise.

While experienced lifters looked more confident and helped each other push more in the gym. Most people use headphones or earphones in the gym and the majority of people are on their phones at every free time they get. If people are not on their phone, they are either zoned out and staring at something random or staring at themselves in the mirror.

Environment 4: Arch climbing wall

The climbing gym showed to be a very tribal environment. There is an unspoken hierarchy assumed by most members at the space. People tend to be more comfortable in company of someone that “knows what they are doing” and find joy in people’s encouragement despite failing at a climb.





Generalisation of observations:

To narrow our study on a more specific context within gyms we ranked our 4 environments based on the richness of the data it provides us. Indoor training gyms like Qmotion and The Gym Group revealed more consistent and insightful data as they are more commonly accessed and relatable to a wider audience. We involved all environments in the analysis stage as some of our observations from them remain relevant and can help inform conclusions about indoor gym spaces. However, coding categories were formulated with stronger relevance to indoor gyms, therefore allowing us to extract the most insightful elements of the outdoor and rock climbing gyms.

1.3 Analysis:

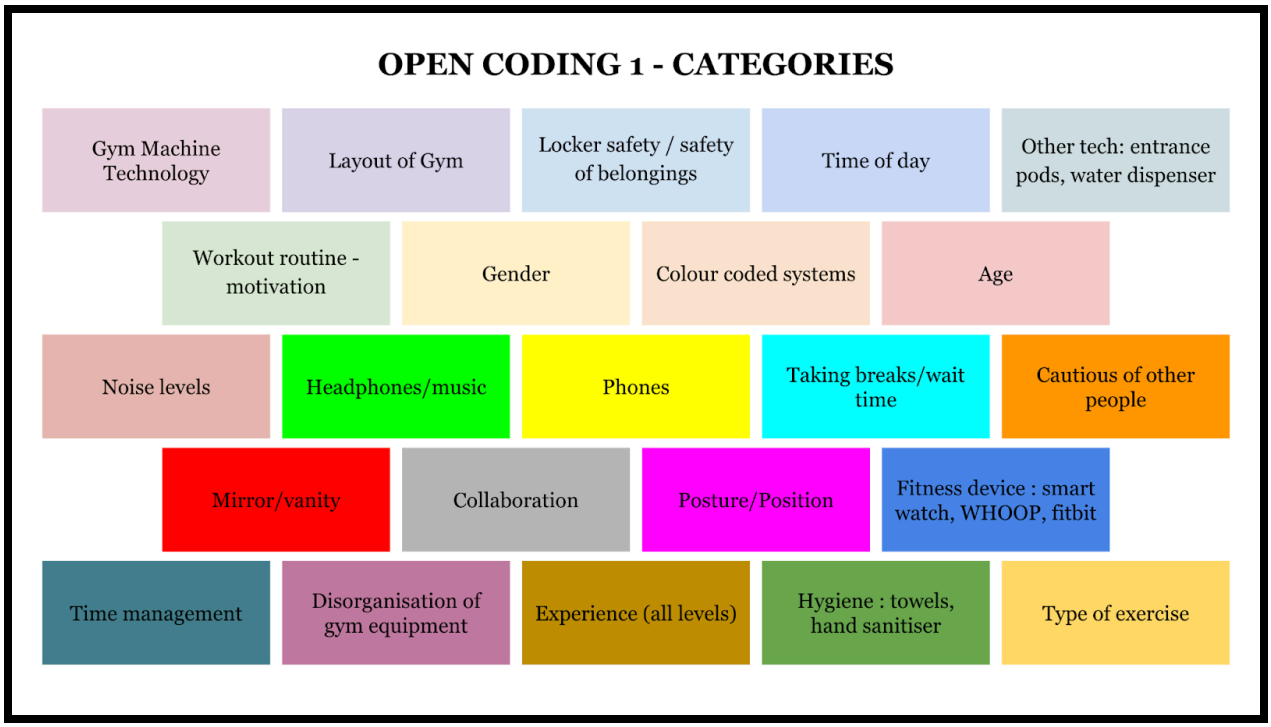
Stage 1: Open Coding

Over a period of 3 weeks all 4 members of the group observed 4 different environments. After the first 2 weeks we all noted and typed up our observations and began the process of open coding. This selection of data has been collected over many different days and we have named this set of data “Data Collection 1”.

Shown below are the colour coded categories, which we used to code our first bulk of data in all 4 environments. View appendix 1 - data set 1 to see our open coded analysis of our first 2 weeks of observations.

What were the key categories that framed our observations?

From the 23 categories identified below, certain categories were much more prominent amongst all of our observations from the varied environments. The layout of the gyms across all 4 environments had an influence on how people distribute themselves based off of visibility. Intensive and high experience exercises are often centred to direct vision as opposed to treadmills which were mostly cornered against walls. This influenced how people with different experience levels maneuvered around the gym. Those who came alone and seemed to be beginners stuck to the edges and hidden parts of the gym as opposed to those who are regular members that walk more confidently and with groups, which is where we saw more collaboration. Time related categories such as management and wait times were heavily influenced by each individual's personal exercise goals and this created bottlenecks around popular machines and areas. People often awkwardly wait around until it's their turn, unsure what to do as an alternative, making their duration of stay less efficient. The lack of assistance without having to ask for it created a sense of anxiety and cautiousness amongst those who are unfamiliar with this environment.



Stage 2: Axial Coding

Part 1

This stage consists of two parts , the first of which involves grouping the previous categories from our initial open coding into similar family groups that relate to each other. This is shown in the table below.

Structure and organisation	• Layout of gym • Organization of gym equipment • Colour coding • Noise levels	**Time and planning**	• Time management • Time of day • Taking breaks / wait time
Gym utilities	• Gym Machine technology • Locker safety / safety of belongings • Other tech: entrance pods, water dispenser	**Personal Devices**	• Headphones/music • Phones • Fitness devices: smart watch, WHOOP, fitbit
Interpersonal Interactions	• Workout routine - motivation • Cautious of other people • Collaboration	**Hygiene**	• Towels, hand sanitizer
Demographic statistics	• Gender • Age	**Personal goals**	• Mirror - vanity • Posture/position • Type of exercise • Experience

Part 2

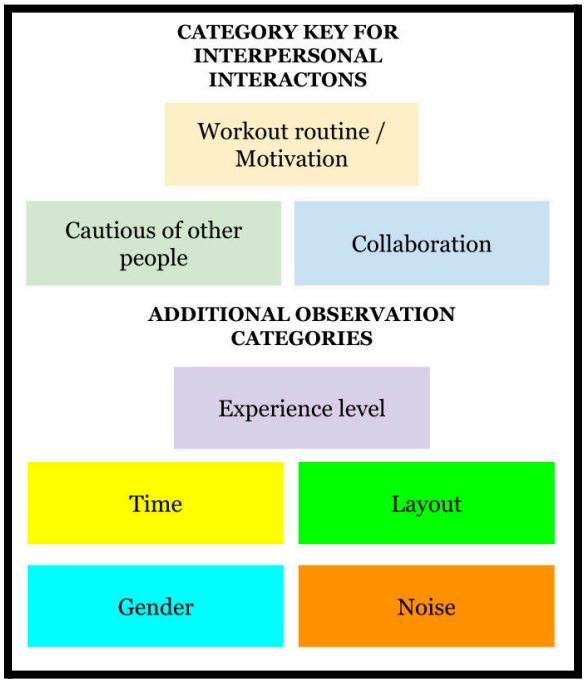
The next step involved organising the axial coded categories into a coding paradigm. We had a vague idea of what our phenomenon of focus would be, however the supporting factors of the paradigm required further observation and data collection.

Revisiting Stage 1:

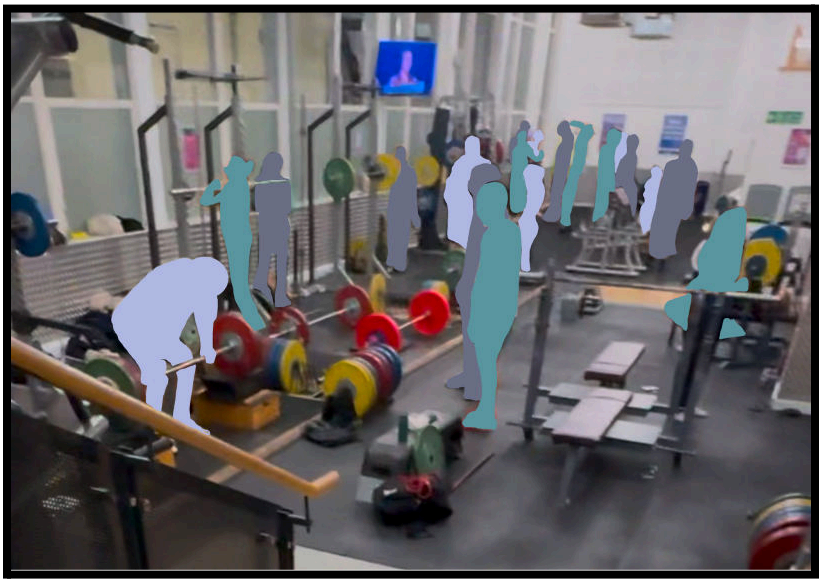
We spent another week collecting more data in our specific environments and putting more focus towards our chosen phenomenon. We then re-coded this data with any relevant coding categories. View appendix 2 - dataset 2 to see our additional open coded analysis of our third week of observations.

Summary of new key findings and coded categories

From our new observations we have confirmed that our key category will be interpersonal interactions with a clear idea of what the supporting factors will be. Collectively, we have seen that experience levels affect your routine and how you interact with people at the gym, whether positively or negatively. Additionally, the time affects the noise levels which affects the part of the gym a user uses. Demographics were also seen to play a part as for example, women tend to be more reluctant to ask strangers in the gym for support.

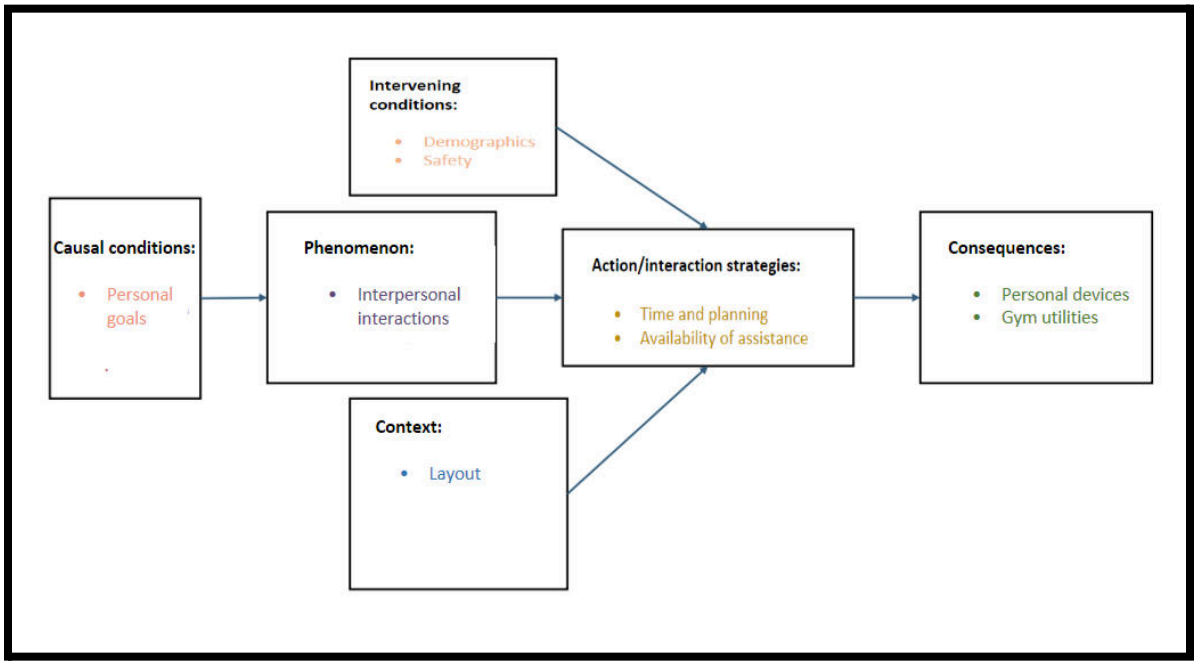


QMOTION HEAVY LIFTING AREA



The Coding Paradigm

After further coding and data analysis we had more data to support our coding paradigm with new categories being introduced after further focused observations.



Selective coding - storyline

Core category: Interpersonal interaction of people at the gym

Gym goers face problems regarding interpersonal interactions. These have different impacts depending on whether you are a beginner or loyal gym member and can affect your routine and the way you collaborate with others. This is driven by **causal conditions** from types of exercises to your own personal reasons and goals for going to the gym. The factors shaping how this issue manifests were found to be the structure and organization of the gyms. This includes the layout, noise levels and the gym equipment themselves. The main **intervening conditions** are safety and demographics. The way people interact with each other varies with demographics. How users should attempt to overcome this phenomenon is by the assistance of trainers and your own time and planning. This ranges from the time of day you use the gym to how you manage workout time and breaks. The **consequences** are that people tend to use their personal devices more.

What is our research question?

How can we introduce technology in the gym to embrace collaboration and reduce anxiety ?

1.4 Problem Definition

Behavioural framing is a design rationale where systems integrate notions of “space” and “place” to shape a collaborative environment. In a gym, users are physically located in space however they act in “place”; a setting defined by cultural and behavioral expectations. Understanding emergent patterns of human behaviours in a “place” allows us to draw correlations to the affordances of the physical space. The design of spatial systems should consider the cultural and behavioural expectations of the place and how it shapes interactions rather than just the spatial organisation and layout. *(Harrison and Dourish, 1996)*

Understanding the phenomena of gym culture will help frame a stronger degree of connectedness to the environment. The aim of our design solution should be to enhance the user experience where a new system should provide cues that initiate more collaborative social behaviour in a spatially organised environment like a gym. *(Harrison and Dourish, 1996)*

Overall the problem is that users are not encouraged to use the gym space to its full advantage. The gym is a place with the unique opportunity to collaborate but this has not been embraced. We want to design a technology to embrace collaboration in order to reduce anxiety.

Part 2: Design

2.1 Concept & Justification

Literature justification

Higher levels of experience in muscle building exercises reveal to show much more narcissistic behaviour explaining why findings concluded less people approaching the deadlift and free weights section unless they are proficient. Increasing regularity and satisfaction of experience at the gym is what will benefit the business financially and the user emotionally. Studies show how dissatisfaction of bodily progress is a major catalyst of users discontinuing their memberships. To avoid activating the catalyst, it is important to create a design that allows a shift from gym-shy to gym-friendly attitudes, where users are guided through a better workout experience to boost their confidence. *(Stewart and Smith, 2013)*

Gym culture has evolved into a gendered space due to female discomfort around male centric machine use. Providing gender specific gyms may be beneficial for some, however promoting segregation as a permanent solution rather than attempting to promote collaboration and reduce social discomfort is feeding into this culture. *(Rizzo, 2017)* Getting a gym membership is a paid experience therefore users should be rewarded with an inclusive service delivery process that amalgamates communicative and interactive systems for more freedom regardless of gender. *(Giersing et al., n.d.)*

A study on Multi-user table tops for an interactive system called “TouristPlanner” brought attention to the social discomfort associated with multi-user environments and tension in unwanted interactions. Different user groups approach and distribute themselves in shared environments differently so the design must encourage coherent usage of the system to make it more collaborative in a non intrusive manner. *(Marshall et al., 2011)* A study on interactions at museum exhibitions highlights the importance of increasing dwell time at installations by facilitating multi-party interactions. Multi-party and modal system designs promote collaboration and avoid restriction of co-participation which is essential in gyms. *(Heath, Lehn and Osborne, 2005)*

Research has shown collaboration plays a crucial role in energising and motivating people when doing any kind of exercise. An article published by the School of Anthropology & Museum Ethnography stated how parkrunners who attended with family and friends felt more part of the community which impacted their emotional experience. This resulted in them enjoying their running with increased energy levels that correlated to highly increased running rates. *(Davis, 2016)*

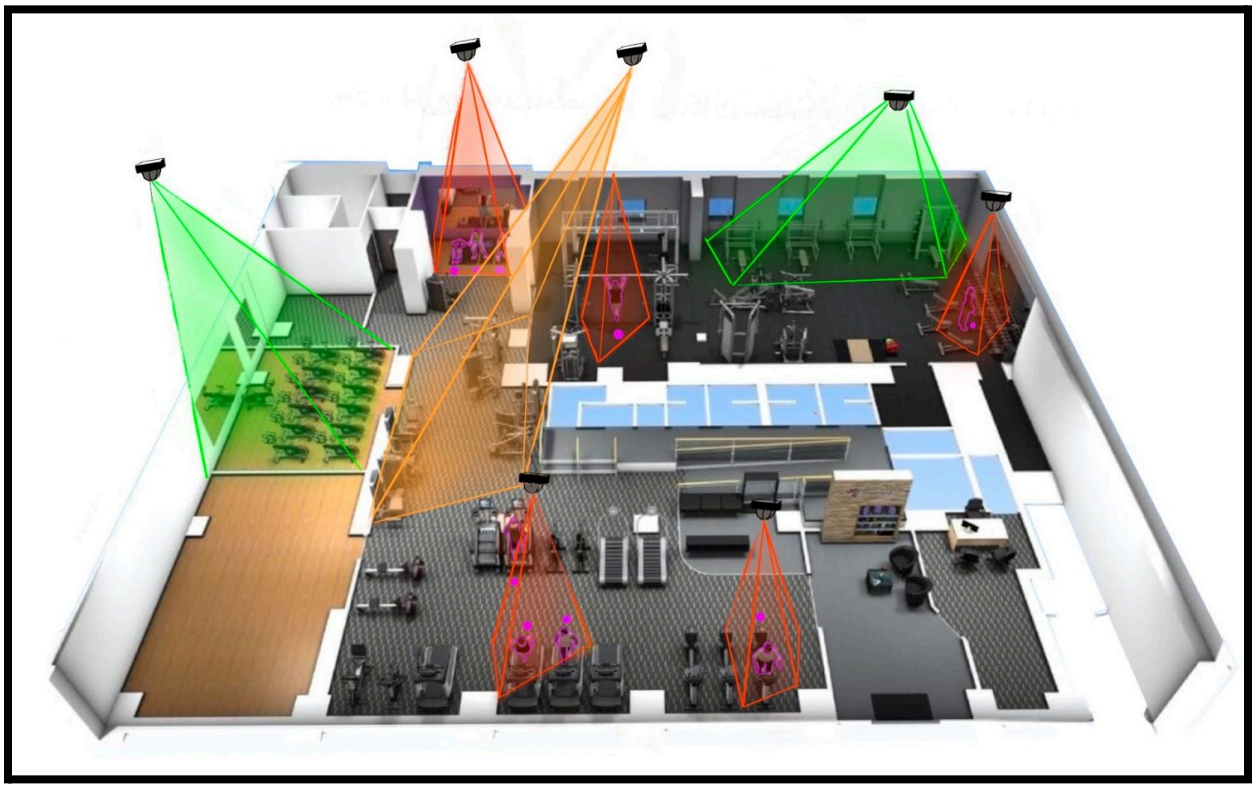
THE CONCEPT

Our design takes advantage of the membership cards that are currently only used to enter and exit the gym. Users can extend the use of their NFC cards to interact with digital touch pads that

can be accessed by all members. This digital system enables users to personalise preferences and curates an optimal and efficient routine based on machine availability and desired time frames and allows collaboration with other gym users. Live data is fed into the system via ubiquitous camera and motion sensor technology that maps the environment and updates the system to the touch pads and to all personal wearables and mobiles, becoming a multimodal design. This accounts for different levels of the gym, availability of exercise areas and machines, noise levels and creating heat map distributions. Data amongst all members is assessed and shared in an encrypted manner within the system database to create collaborative member matches based on routine similarity and experience levels. It allows the user to collaborate with people with less social discomfort and invasiveness as members can customize their preferences on becoming a buddy or not. The aim is for the user to build a streak of consistency where their card marks their loyalty. The multimodality of the system communicates to the user through more than one device for seamless experience and amalgamates both ubiquitous and manual data inputs.

Ubiquitous camera and motion sensor technology

The “Just Walk Out” technology employed in cashierless stores such as Amazon Fresh, utilises computer vision, deep learning via processing and analysis of photo and video data. The environment is functional as a multimodal system, installed with multiple AI technologies where customers interact with their own personal devices that connect to the store's app and live data. Their purchases are recognised by this technology and synchronised with their devices, making their experience effortless and automatic. *(Eszter Szabó-Szentgróti, Szabolcs Rámháp and Petra Kinga Kézai, 2023)*



To ensure our system runs more seamlessly we have drawn inspiration to use such technology to live track the flow of users and avoid a bottleneck in everyone's routine across the gym. This is how the entire gym's availability mapping and routine planning will be executed. Machines and areas that are unused will be recognised and updated to the systems database which will then be synchronised to all personal devices where all users routines will be updated to guide them to the available exercise. *(Eszter Szabó-Szentgróti, Szabolcs Rámháp and Petra Kinga Kézai, 2023)*

Norman's Design Principles

We have changed some aspects of our design to make them align with Norman's design principles. For example, in terms of affordances and signifiers, we will show to the user "tap your card" instead of just "tap" as this might confuse the user into tapping the screen. We have also used sensors to make sure the mapping of the gym is accurate and easy to follow. In terms of feedback, the ability to connect to your smartwatch allows for different modalities of interactions such as flashes or vibrations.

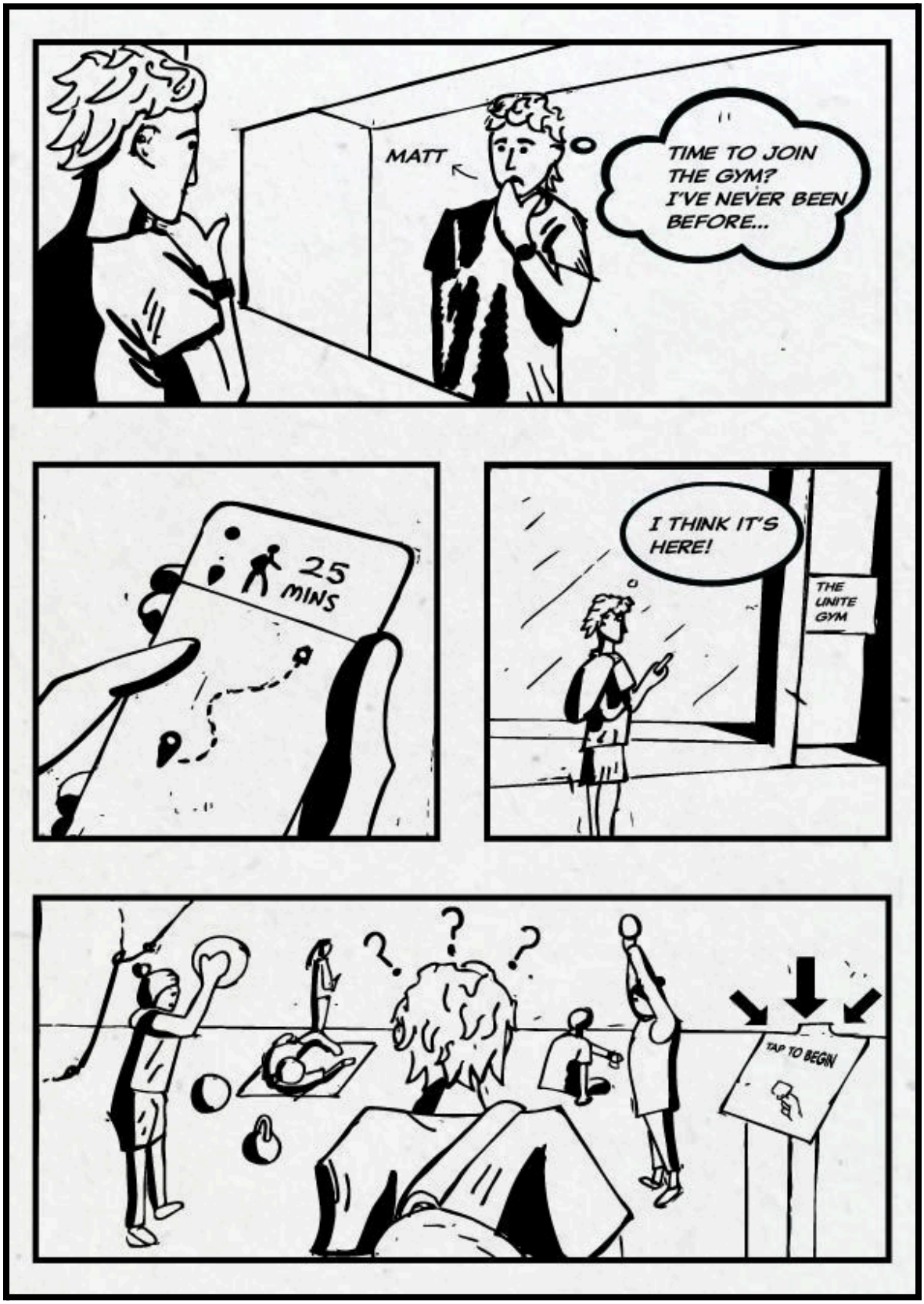
Phenomenology

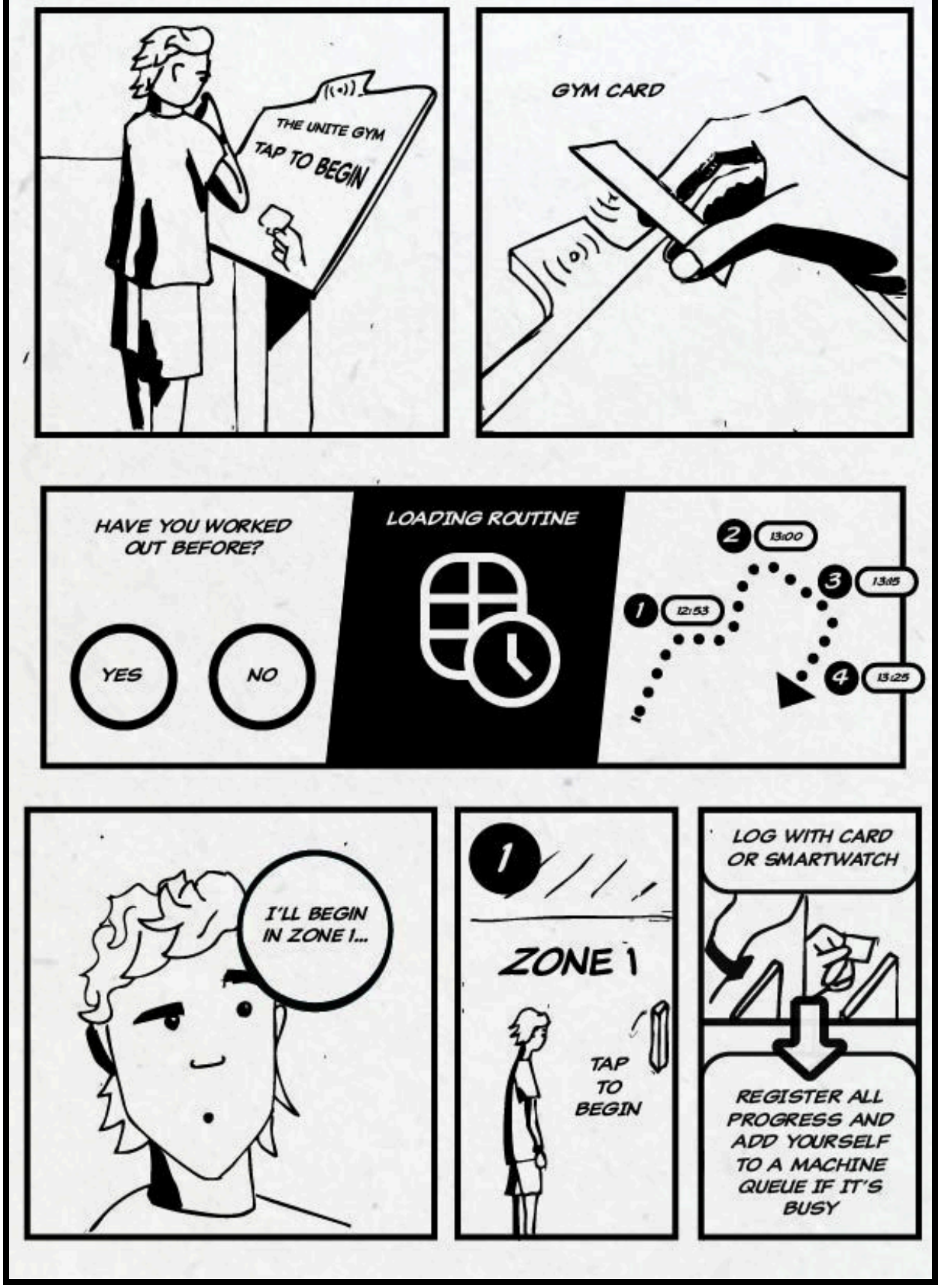
When looking at the phenomenology of gyms, the understanding of space is important. As "although space can be given a quantitative structure, its basic structure is entirely qualitative". Your experience level draws you to certain sections of the gym and makes you avoid others. We kept this in mind and implemented this into our design, seeing the correlation between a user' personal routine and the gym layout. Additionally, users want to be near people and collaborate or want to isolate themselves. Since we are encouraging collaboration, our matching system allows you to match with users in a less intimidating way. *(Healey et al., 2007)*

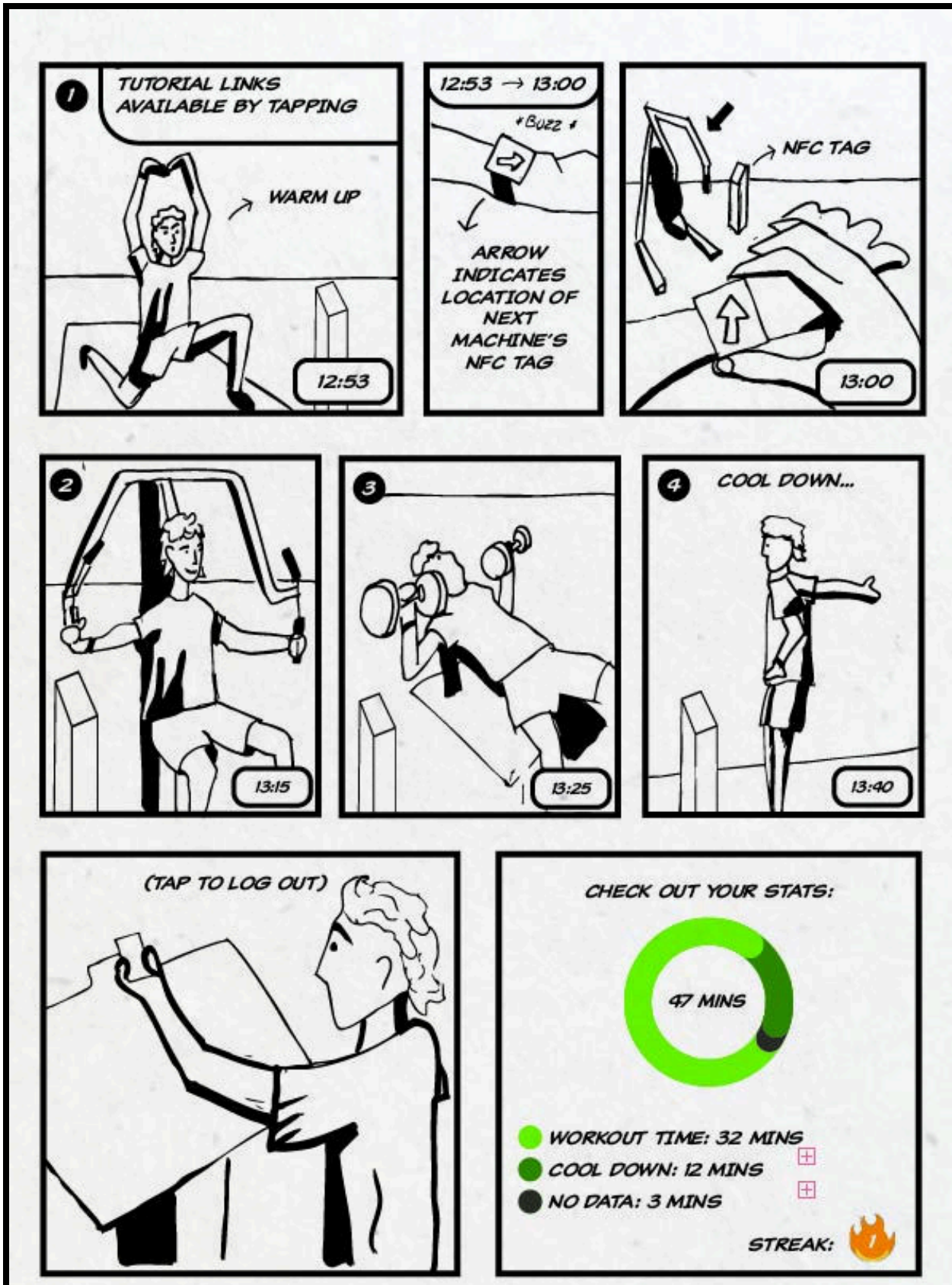
Colour Theory

Colour theory and psychological functioning are highly interlinked where they can be combined to instill specific emotions in different environments. The aim is for our users to feel safe and less anxious and eventually more confident in a gym environment, resulting in more collaboration and consistency. Studies have shown that blue activates the melanopsin photoreceptor system resulting in higher levels of alertness, performance and sustained attention. Blue system designs often accompanied with green have shown increased quality and trustworthiness appraisals, hence our chosen UI colour scheme. *(Elliot, 2015)*

2.2 User Experience Storyboard

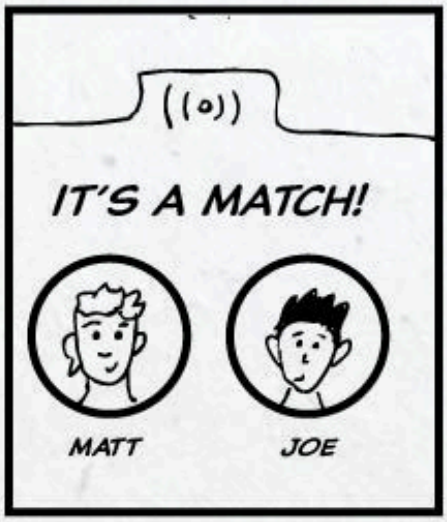



THE UNITE GYM
TAP TO BEGIN
GYM CARD
HAVE YOU WORKED OUT BEFORE?
YES
NO
LOADING ROUTINE
1
12:53
2
13:00
3
13:05
4
13:25
I'LL BEGIN IN ZONE 1...
1
ZONE 1
TAP TO BEGIN
LOG WITH CARD OR SMARTWATCH
REGISTER ALL PROGRESS AND ADD YOURSELF TO A MACHINE QUEUE IF IT'S BUSY


1
TUTORIAL LINKS AVAILABLE BY TAPPING
WARM UP
12:53
12:53 → 13:00
Buzz
ARROW INDICATES LOCATION OF NEXT MACHINE'S NFC TAG
NFC TAG
13:00
2
13:15
3
13:25
4
COOL DOWN...
13:40
(TAP TO LOG OUT)
CHECK OUT YOUR STATS:
47 MINS
WORKOUT TIME: 32 MINS
COOL DOWN: 12 MINS
NO DATA: 3 MINS
STREAK:
1

THE NEXT DAY





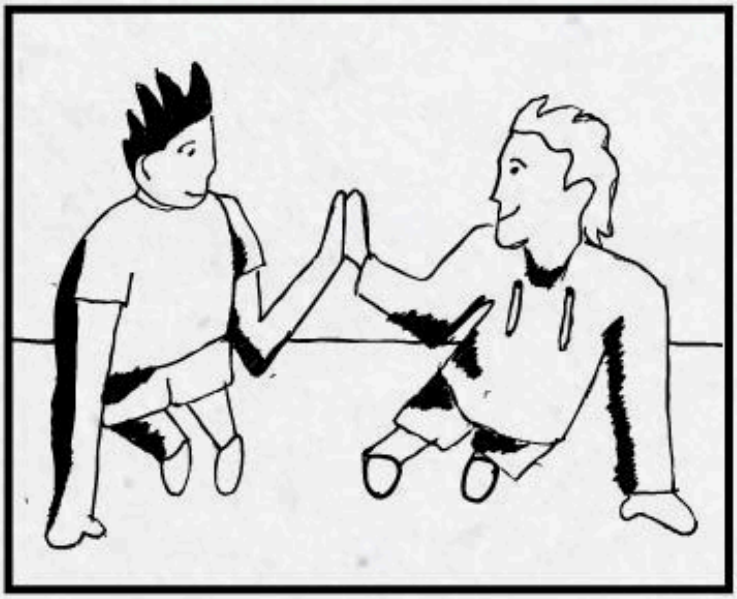
THE DATA CHECKS THAT MATT AND JOE ARE ON THE SAME LEG ROUTINE.

JOE IS AT AN ADVANCED LEVEL AND CAN TAKE THIS MATCH AS AN OPPORTUNITY TO LEARN TO COACH.

MATT IMPROVES FASTER AND HAS SOME GUIDANCE!

MACHINE TIME IS OPTIMIZED TOO!





CHECK OUT YOUR STATS:

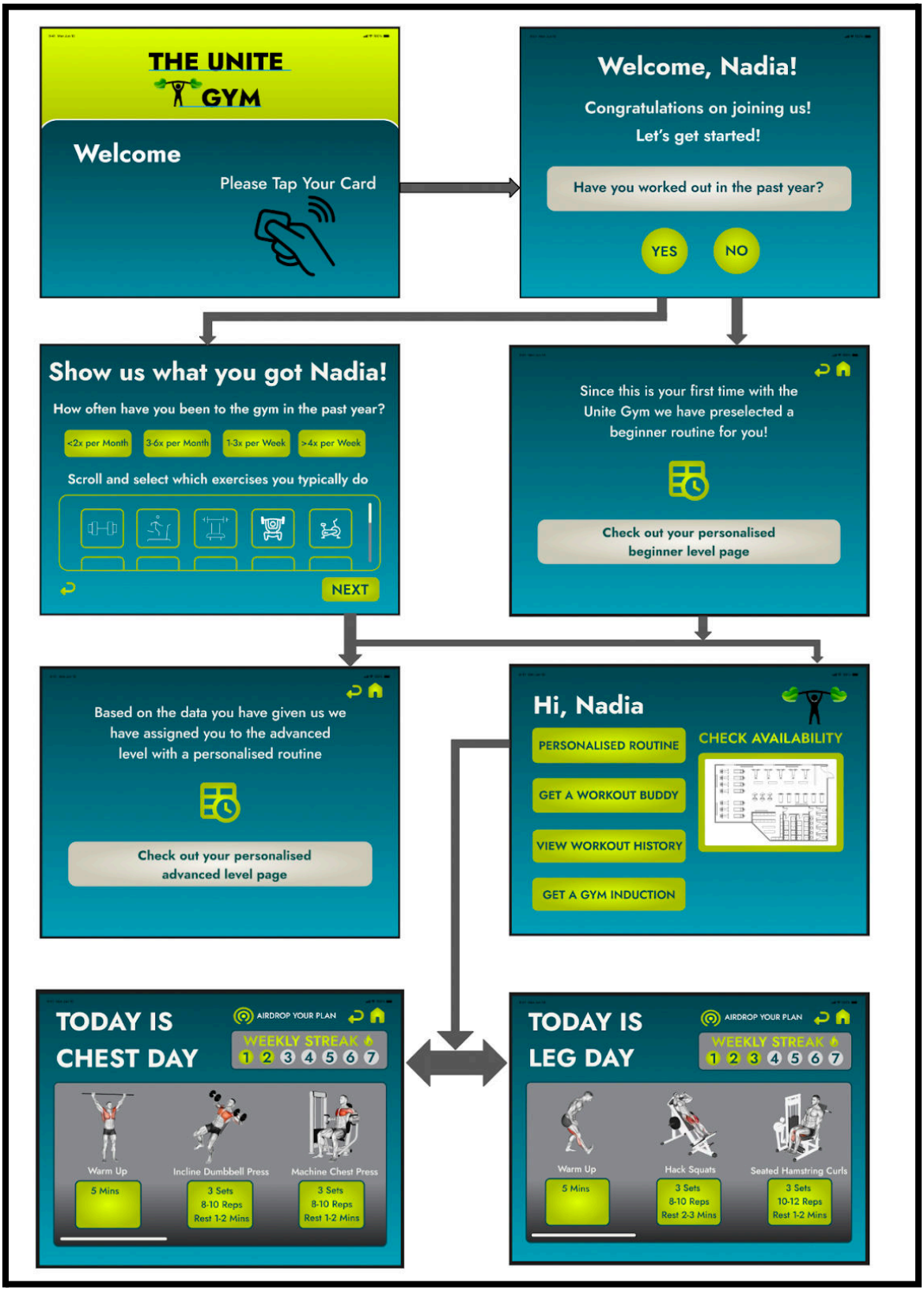


WORKOUT TIME: 45 MINS
COOL DOWN: 15 MINS
NO DATA: 3 MINS

STREAK: 2

THE END

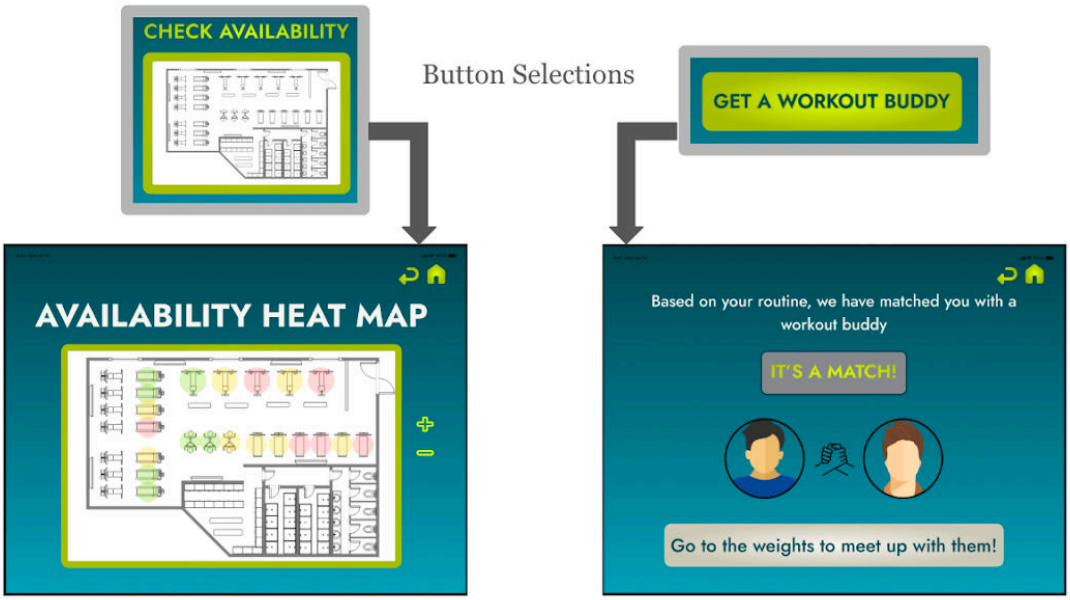
2.3 User Interface Wireframe

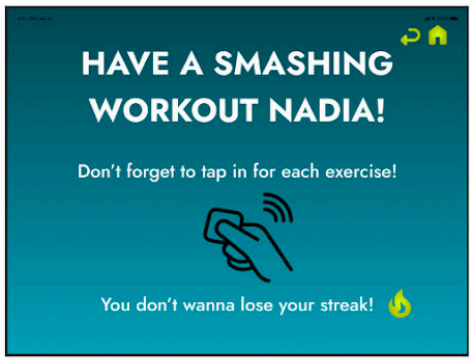


Button Selection



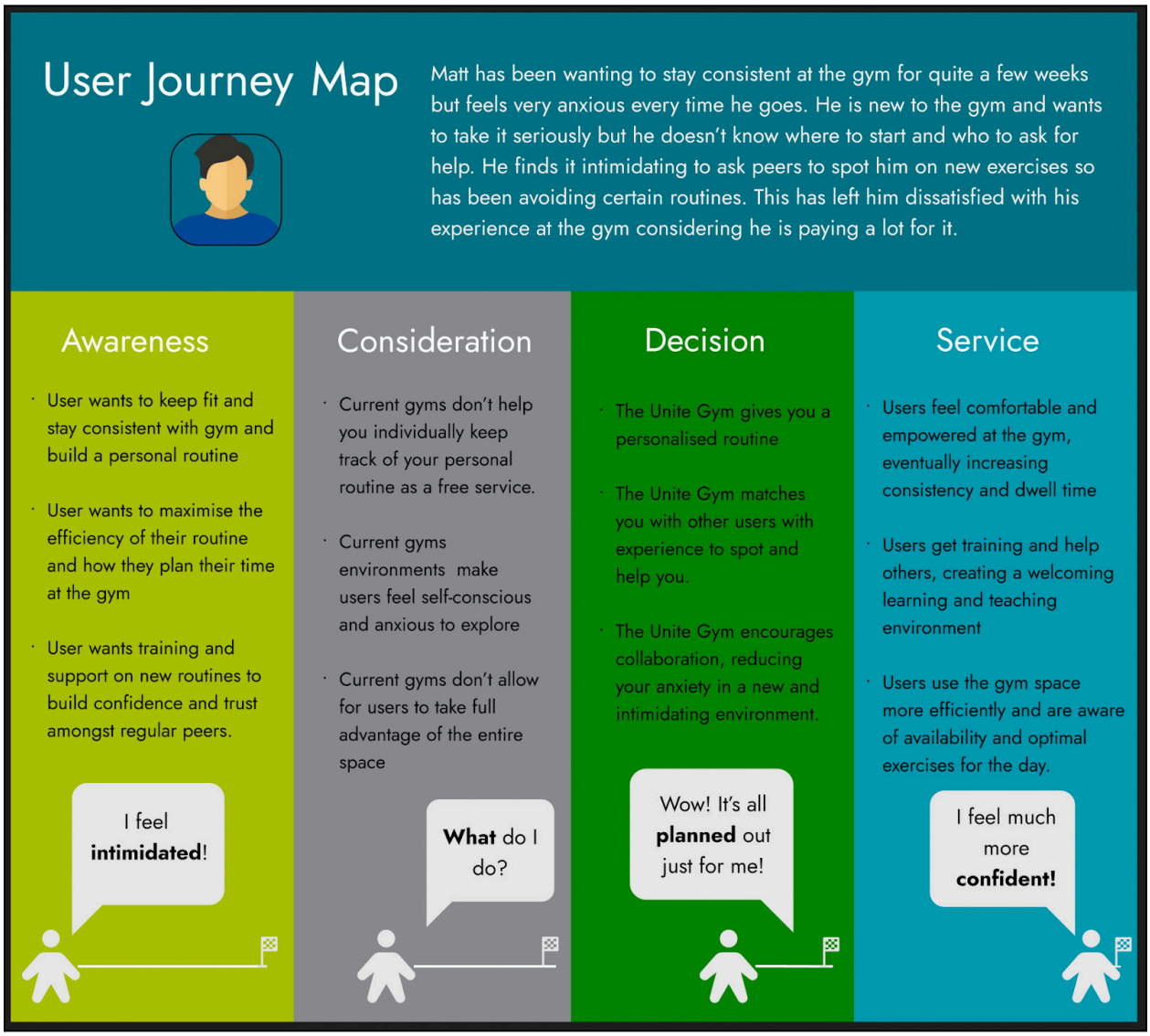
Button Selections





Link to Figma prototype: https://youtu.be/WZpozv4bIo4

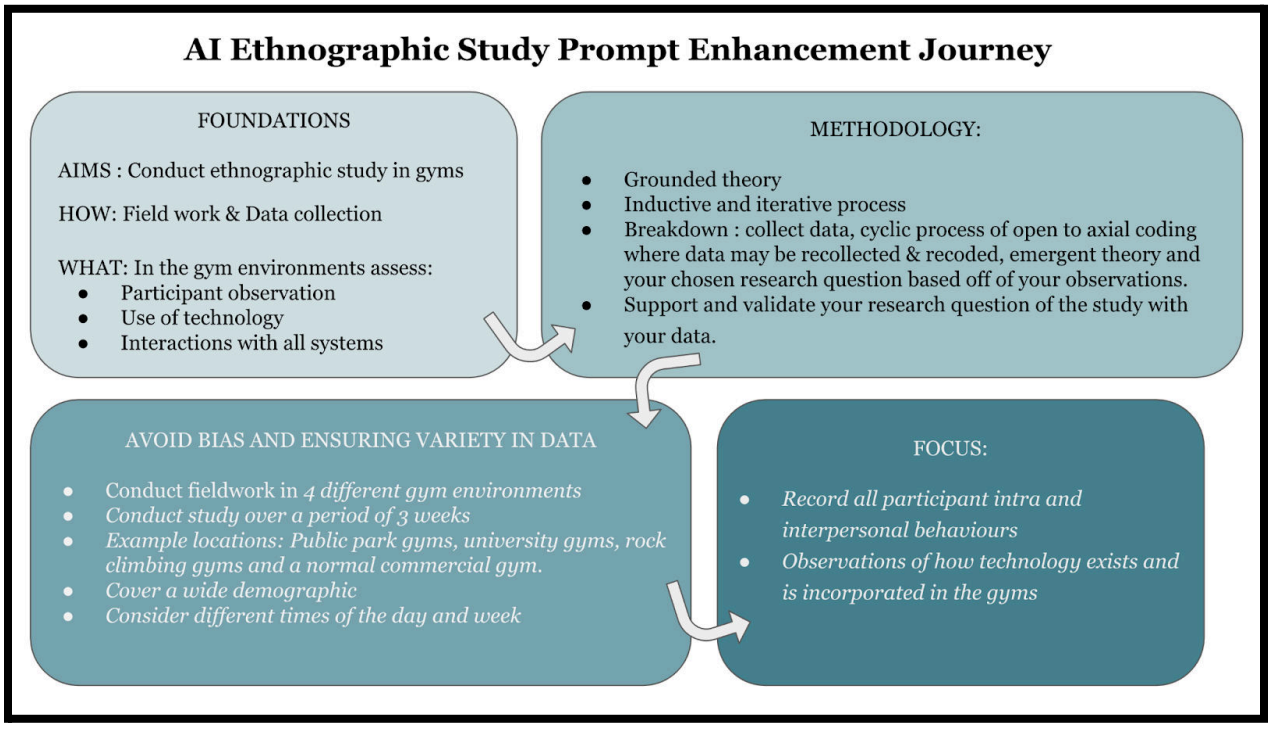
2.4 User Journey Map



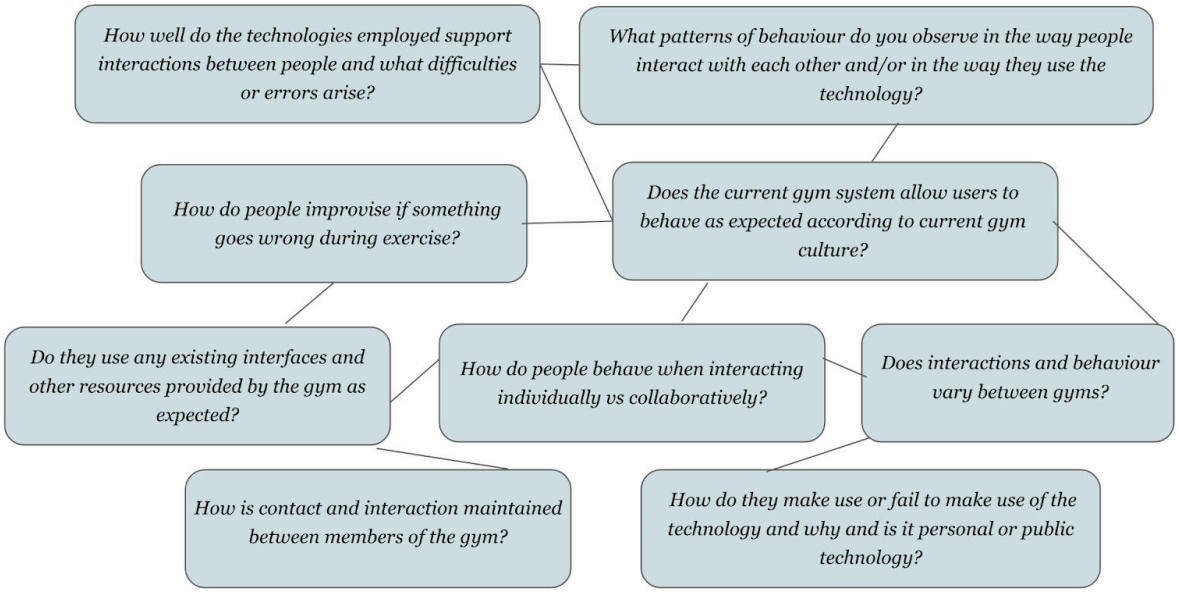
Part 3: Analysis

3.1 AI Prompt & Response

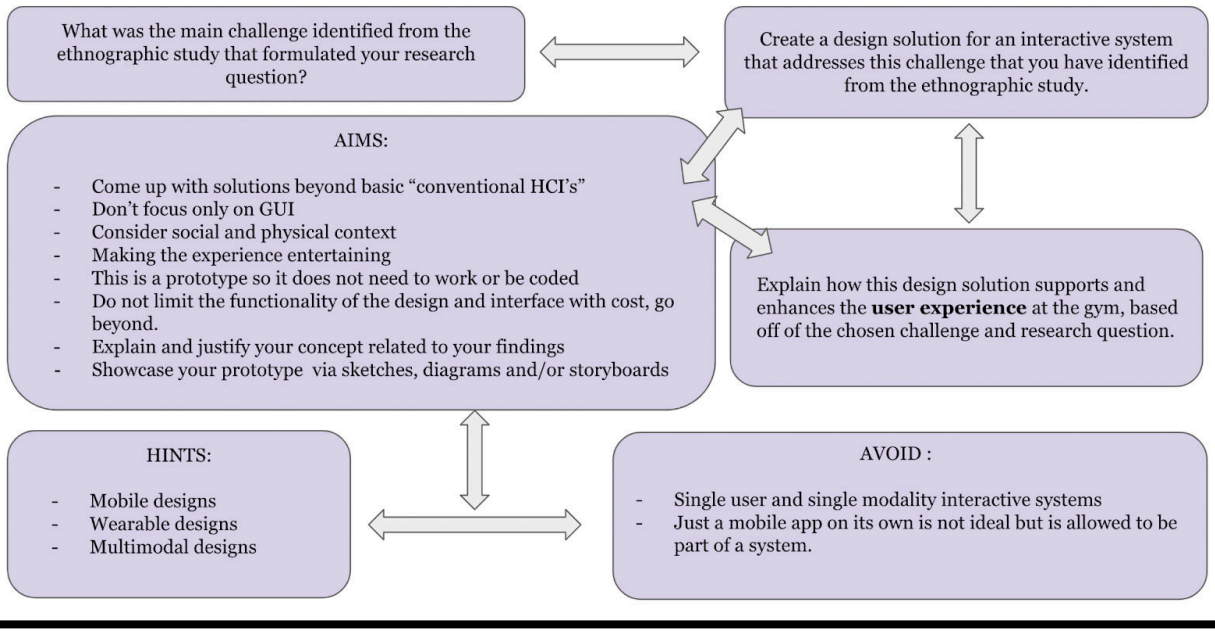
We used ChatGPT 4_o as the AI tool to perform an ethnographic study and to develop a consequent interactive design solution addressing the key challenge and research question it had concluded. We had to curate informative prompts that explained key considerations when AI formulates its response. The diagrams below show how we mapped out and consolidated the essential factors that frame this task. The aim was to create prompts that give enough guidance to form a fair comparison in order to compare and contrast findings and solutions, however not to provide a direct solution or research question. Since we followed the grounded theory methodology which is inductive and iterative we wanted to assess how AI could perform qualitative data fieldwork and analysis with the same method. The various boxes and flow mapped structure of the diagrams below show how we refined and combined our draft prompts to create two cohesive instructional prompts; one for the ethnographic study and one for the design. Entering one prompt at a time allowed for more detailed and informative responses as opposed to combining both prompts into one. View appendix 5.4 and appendix 5.5 to see the full AI response to our prompts.



AI Ethnographic Study Prompt Enhancement Journey



AI Design of interactive technology Prompt Enhancement Journey



ChatGPT 4.0 divided its findings in four separate categories. In the first category, Social Interaction & Technology, it assesses the dependency of technology in different gyms and the amount of social interaction where it acknowledges that commercial gyms had minimal social interaction. In the second category, Technology as a Motivator vs Isolator, it finds out how technology can act both as a motivating and isolating factor for the gym goers, where it mentions that fitness apps replaced the need for human interaction in the university gyms. The third category, Adaptation & Improvisation, studies how technology or lack of technology can insinuate people to adapt and improvise their workouts. And the last category of the study, Mismatch Between Technology and User Needs, focuses on how some technology is underutilized in the gym and doesn't align with what people actually need.

Based on the prompt, ChatGPT 4_o has designed an AI-powered wearable system "PulseBand", designed to enhance social interaction, motivation, and adaptability in gym environments. It features AI-driven proximity matching to connect users with similar fitness goals, haptic feedback for real-time coaching and form correction, and shared AR projections to display progress on gym walls. Additionally, it integrates with gym mood lighting to adjust ambiance based on collective energy levels. PulseBand is adaptable across various gym types, providing personalised feedback and interactive guidance. This design aims to bridge the gap between individual performance tracking and communal workout experiences, fostering engagement and motivation.

3.2 Comparison to our findings

Comparing the ethnographic study and observations the AI created to the one we conducted, there are similar themes but with some key differences. Themes included were using technology as motivation and the use of personal devices. These themes were tied to both the physical and emotional isolation which was seen as significant in both studies. From both observations, this is specifically more prevalent in commercial gyms. An interesting insight the AI gave was on observing what happens when gym technology fails which is not something we had the chance to observe for our study. However, the AI' observations started going in a different direction as it started looking at virtual reality such as the "immersive VR treadmill experience" and use of augmented reality for route planning inside the gym which is not something we have ourselves seen.

When looking at the design the AI created, whilst there were some intriguing features it did not seem as feasible. Similar to our design, it encouraged social interaction by matching with people in a similar process to ours, although it suggested using "AI-Powered proximity matching". Whilst it included the use of wearable devices, this is the only type of technology the AI used, whereas our design included pervasive technology as well. The AI' design went in a different direction from ours as it provided additional features such as the ability to change the physical

environment by changing the lighting. From this it expanded its scope, looking at “multi-gym integration”. Although we integrated a way for users to continue from their previous gym level, we did not go down this route. We decided instead to take a more focused approach, solely looking at commercial gyms as this is the environment for which our design would have the greatest impact.

Other similarities included milestones, a "community" display and providing the user with a way to keep and track workout data and communicate this back to you. Additional features for user feedback by the AI included audio prompts and haptic feedback. Whilst we also used haptic feedback in terms of vibrations from your smartwatch, we only decided to use this for reminders and notifications. The AI suggests using it for form correction as well. Whilst we mentioned stretching at the beginning of our observation, this was not part of our key categories for our axial coding so this aspect was lost.

The biggest difference is the overall experience, as the AI's design was only focused on paired workouts, not on user’ individual routine as well. This means a more limited experience as the user can't change settings based on their previous experience, can’t create a personalised routine specific to their experience level and there is no feature for time management. These 3 features were key in our design and the fact that this was missing in the design the AI provided shows that it took a different creative direction.

In conclusion the AI stuck to the key theme of interaction but strayed away by adding technical features that were too complex and would be harder to integrate. From observing the use of immersive and AR gym technology, which is a more extreme example, to implementing these into the design, this felt less genuine and more of an excessive approach. Whilst providing observations we might not have thought of and adding useful features, our analysis and design overall feels more relatable and practical for the use case of members of commercial gyms.

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5. Appendices

5.1 Data Collection 1

QMotion Data Collection 1:

MONDAY : 03/02/25 (1PM)

- Card access machine
- Up the stairs to the right are changing rooms (one water fountain)
- Locker rooms female (some people bring padlocks and some leave their belongings unlocked)
- Private hire lockers have q-motion padlocks
- Phone charging cupboards to place phones during exercise
- 1pm Monday : not too busy
- People on their phones or have headphones on whilst on the treadmill

FRIDAY 07/02/25 (11PM)

Female gym area

- Friday 11:07 (announcement made about quiet hours)
- Not allowed to drop weights or make loud noises
- Most people have headphones on/ listening to their own music
- If they are on a break they will either sit listening to the music or look on their phone
- Very minimal people to people interaction
- Girl on the stairmaster has the screen showing the calorie count but has her headphones on and is looking at her phone . She isn't holding onto the handrails and is using her phone with a single hand
- The girl on the cycle stand has headphones on but isn't using her phone so she is holding the hand rails
- Another girl is using phone and listening to music not holding handrails
- Girl who is doing hip thrusts is sitting down for a while looking around taking a break

FRIDAY 07/02/25

- Different types of exercises require different break periods
- Mostly everyone has with headphones, AirPods or earphones in .
- Anyone in a break position holds their pose for a few seconds to mins either staring into space or using their phone
- When people are choosing what exercise to do next they often walk around holding a water bottle in their hand
- People are always on their phone in the breaks they move to a corner of space no one is using and isolate themselves on their phones
- Most people hold their phone single hand when they are doing an exercise or holding a water bottle on their other hand
- Hold phone with both hands when sitting or away from equipment
- A girl came to a leg raise exercise and sat down and paused. She then checked the weight she was on and looked around without adjusting it and starting doing the leg raises. As she was doing them she was listening to music and anxiously looked around doing a few reps and then holding a pause to take a break between sets.
- She looks around after doing a few more then increases the weight and goes on her phone.
- People often sit on the equipment for long periods to mark their turn so no one approaches them between sets
- I was standing near an exercise bench on my phone so a girl came up to me and asked if I was using it indicating that pausing for long periods of time near equipment marks territory over the equipment
- No one was talking to each other only two people were having a conversation and they came to the gym together
- I was sitting on an exercise block and a girl asked me if she can put her phone there next to me while she used weights in front of me
- People often scatter their belongings near the area they work in on the floor (included water bottles and phone)
- One girl was laying on her back on the floor with her hands in the air on her phone
- The two friends who came together were talking whilst on adjacent pieces of cycle equipment
- One girl was doing an exercise facing the wall instead facing the way that had more people and equipment
- People like to workout in front of the big mirror to watch themselves do the exercise
- When setting up for an exercise people place their weights down and stand with their phone for a bit

FRIDAY 07/02/25

- One girl started an exercise and then paused midway so when I checked she had her phone facing up in front of her and she was following a TikTok tutorial on how to do an exercise with dumbbells. She then pauses it and sits down and goes to grab her water bottle
- She then did her own new exercise but was more hesitant on posture as she wasn't following a video so her movement would change frequently
- She then stops the exercise and starts searching up on google different dumbbell exercises and the website shows a wrong and right posture image
- She then sits down and begins to do some stretches instead
- She then reattempts another exercise and is a lot more consistent with it

12 people in the female gym at 11:34

- 5 using machine based exercises
- 3 doing standing dumbbell exercises
- 3 on their phone
- One setting up bench and looking in mirror with heavy dumbbells

Main gym area

- Only men in front of the dumbbell area
- Lot less women
- Lots of people using the treadmill
- Again if people aren't using equipment they are on their phones standing
- A lot more conversation is going on between people
- They behave a lot more aggressively
- More high tech equipment

Tuesday 11/02/25 - 3:37PM

- main section completely male dominate
- Give u a look as you walk in
- Intimidating
- Many are crowded around the mirror
- More conversation is happening
- They tend to sit / lean on equipment when taking a break to make "territory"
- Leave belongings to show they are using equipment

Female Section:

- lot busier in later parts of the day
- Friends are doing the exercise together in sync
- In female section u find more people stretching before exercises
- Girls regularly clean up equipment after themselves with wipe and sanitiser
- When taking breaks they stand much closer to the equipment and look straight down into their phones
- People pick up and carry weights in different ways from the rack
- Not everyone trains with weights specifically in the female section

Male/Main area section

- Much more clustered
- Everyone is bunched around the mirrors
- Guys are helping each other out and pushing to do more
- The girls that do work out here are much more trained
- People sit on equipment with music in their ears and on their phone between sets
- They don't move out of a specific radius from the equipment till they are done

Male/Main area section

- Two mirror sections in the male area
- People come with more friends
- They tend to make much more aggressive expressions
- They walk around a lot more in this space as it is bigger
- Area is divided into different categories
- Deadlifts and bench press/ barbell are in the most evident area more so the first thing you see when you walk in as it has more colour coded weights that mark their weight more evidently
- Cable related machines are all in one square area
- People help each other more near the cables/ more of a shared space
- Treadmill in a more hidden corner with less aggressive people "quiet corner"
- Every time you walk into a space people often tend to give u a direct glance to see who's walking in

Public Park Gyms Data Collection 1:





Outdoor parks - Tuesday at 11am

- Locations: Next to playground, behind playground and next to tennis courts
- Area behind tennis court seems more secluded (have to take a side path to get there)
- Elderly people in main area
- One person had their backpack hanged next to her
- Man stretching + little run around
- No bench around to sit
- One guy switching between multiple machine whilst one woman only staying on pedalling machine
- Woman (alone in the area) started using the phone whilst pedalling then stopped pedalling and just looked at the phone - turned on podcast out loud and continued (holding phone with both hands)
- Man on mat on ground doing exercises with headphones
- Boxing
- Despite there being some machines with adjustable weights - 2 men brought their own dumbbells in shed area - next to main area but not together
- Hanged coats on bar next to them (no designated area to leave coats or bags) + water bottles
- Most people left their coats on
- Friendly/less intimidating atmosphere - fewer people - knew each other
- The elderly people gave me a look
- One waved goodbye to the other and left
- 3 people using “machines”, 2 boxing, 1 with mat, 2 with weights and mat
    -

Saturday 4pm observations

- 2 guys competing at doing pull ups
- Taking turns at machines to see who does it best
- A child playing around with family even though he’s not allowed to use equipment
- Child had to be helped on balance beam as he was too small (toddler)
- Two girls trying the weights for a bit then leaving
- Son and father playing basketball near

The Gym Group Data Collection 1:

Gym Group Layout:

-1 Floor: Machines, Barbells, Free Weights and Benches
0 Floor: Entry
1 Floor: Changing Rooms, Glutes Training Area & some treadmills and cycles.
2 Floor: Some more machines, flavored/ caffeinated water dispenser, cable machines, all types of cardio machines, cross fit area, boxing bag, stretching area.

Entry:

- There's only 2 contactless entry pods, on a busy day (weekday after 5 PM) there are queues to go in, as people one by one write in their codes or scan the QR code.
- People who don't remember the code have to open the gym group app and scan the code, which takes up more time.
- There is only one lift, normally either on -1 or 2nd floor.

Locker/ Changing Room:

- Despite the lockers, people keep their stuff outside or keep it in the lockers without locking it.
- Its always wet and dirty near the sink and hand dryer
- They always run out of soap if you go to the gym at the end of the day.

Glutes Training Area:

- Mostly women are in the glutes training area
- Men are mostly in the glutes training area just to use the cable machine when it is busy on the top floor (where the cable machines are).

Main Gym area:

- Younger people are more often in groups, teenagers.
- Benches and cables are very busy during weekdays and especially after 5.
- People keep their phones, bottles and earphones in front of the bench to protect them from getting crushed by dumbbells.
- Most people use earphones rather than headphones.
- For bench/ sitting exercises people scroll phone in between sets or zone out
- Some people just look at the mirrors or other people working out in between sets.
- Dumbbells are scattered all over the rack, not in their respective numbers.
- Sometimes it is very difficult to find the dumbbells you are looking for.
- Weight plates are also scattered all over the place, very difficult to find 1.25kg & 2.5kg plates specifically, as there are very few and are also very tiny.
- Sometimes very very difficult put barbell spring collar(thing that fixes the plates) to the barbell
- During cable exercise , in between sets, people just walk around the area near the cable.
- Big groups of 4-5 sometimes just hog a machine for a while, practically unusable for anyone else for 30ish minutes
- Very few calisthenics people noticed.
- Mostly men on the -1 floor, very few women seen and they are mostly near the leg machines.
- On a busy day, for the cable machines you have to keep asking people how many sets are left.
- Very difficult to find cables next to each other for chest flys on busy days.
- Also on busy days people ask to jump in with other people on a machine, when the first person rests between sets, the other person does their exercise.
- There are only 2 small sized bars near the cable machine when there are 6 cables, sometimes the cable is free but the required equipment is busy.

Rock Climbing Data Collection 1:

12/1/25

- 10:00 am
- 1 guy → little bag → brush powder, earphones
- → He climbs - fails→ cleans the board → accomplish it.
- → lockers 50% empty, 30% locked, 20% open.
- → 1 blonde guy → long sweatpants 15 mins looking around, does nothing.
- → INFLUENCER → has a mic and camera crew, draws attention| unable to accomplish 4 attempts at the *interactive board, finally completes it on 5th attempt.
- → No one tries the board for the remaining time I'm there.
- High school Kids show up, all in their school apparel. They Are by far the best climbers.
- I teach a person how to take a climb and he accomplishes it; I feel more confident.

MILE END CLIMBING WALL

- Login to the system, watch safety videos.
- Went with friend at 4:30
- Monkey room, training room, sauna, gym, bouldering and climbing room
- Two 10 year olds on private session, they can climb almost any level
- QMUL climbing society shows up and bring good vibes
- They share the a chalk bag between 3
- A guy works out barefoot in the gym section
- We drink cucumber water and member of staff offers to refill
- Teacher explains a climb to me and I improve my mark by a bit
- We exit gym
- I attach climbing shoes to my backpack

5.2 Data Collection 2

QMotion Data Collection 2:

- Evening 5:37 - main section
- Groups of boys are hung around the cables areas
- Later on in the day after uni finishes , friend groups come together so the environment is much louder
- They are all clustered in the middle (cables) and left section
- Girls are more clustered near the machine based cycle/treadmill section
- Girls have headphone in and are much quieter / calmer
- Lots of boys standing around and just talking/ socialising and laughing
- There are multiple machines but not many people use them in the main section
- Three girls came together and started doing the cycle machines together. They were much more comfortable in this main area as they had people to talk to. they were doing the same workout at the same time and next to each other.
- Cables area:
- All boys
- Working in groups of 5
- Pairs
- And solo
- Using the cables to do different types of exercises and helping each other / motivating each other

- Deadlift/barbell area much more open space
- Girls helping girls out
- Boys helping boys out
- In front of the mirror also 5 people were flexing at the same time reminding were sat on a bench looking at themselves
- With leg machines people are much quieter and to themselves
- Any weight lifting involves more collaboration especially deadlift/barbell area
- Much more noise in this deadlift/barbell area
- Female gym 5:47
- Extremely quiet
- Girls use the treadmills and cycle machines much more than boys
- No conversation at all in female gym
- Girls often face away/ towards the wall and look down at the floor or their phone
- Boys are more loud and carefree
- Only ones who come solo are quiet
- Girls sanitise equipment
- Much more collaboration with weights / dumbbells compared to machine assisted exercise
- But all of them are much more experienced and bash the weights to the ground
- They don't look like new comers and are more comfortable in that space
- This is a high experience ranked area

Public Park Gyms Data Collection 2:



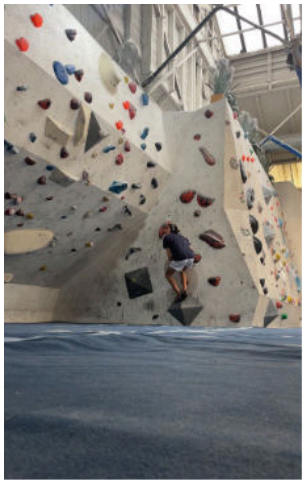
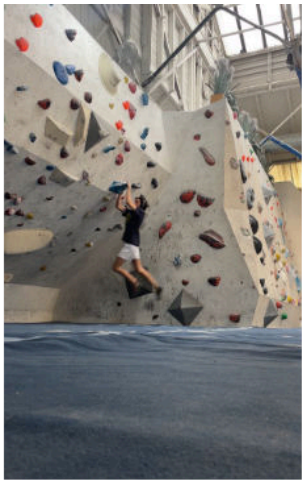
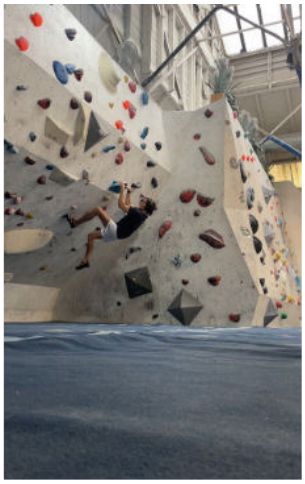


The Gym Group Data Collection 2:

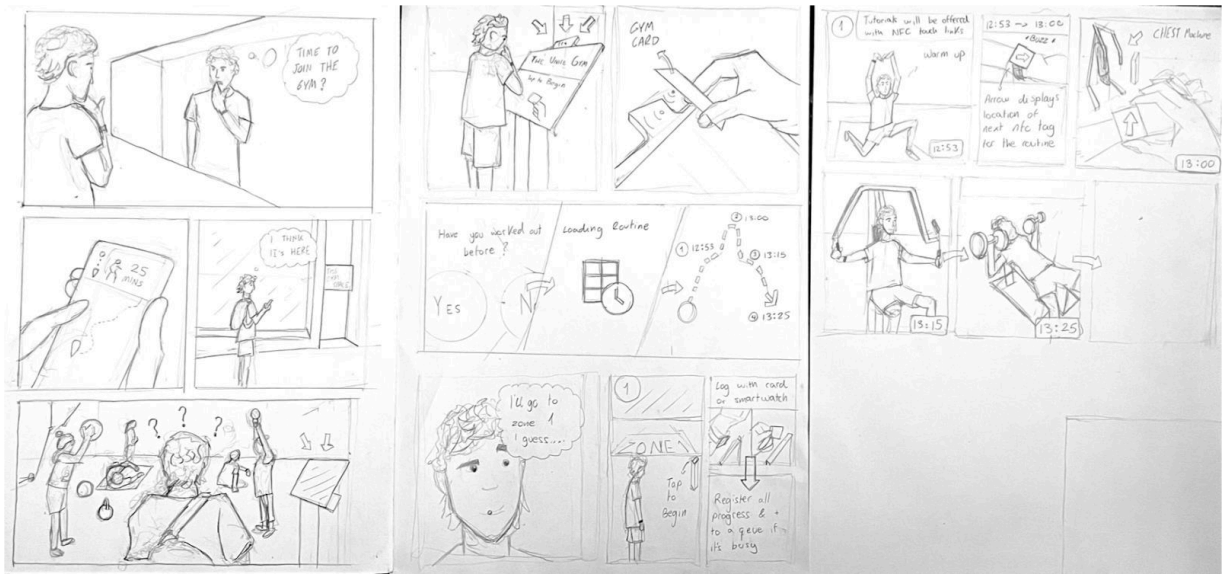
- More experienced friend guides the inexperienced friend about how to do an exercise
- Beginners underestimate the weight when starting a new exercise, one guy had to decrease weight twice on the cable machine before finding a suitable weight to do lateral raises.
- Two experienced lifters when working in collaboration support each other, help each other grind that extra rep at the end of a set.
- Also experienced lifters help each other get free weights up when doing a pressing movement like bench press.
- Amateurs sometimes just for the sake of showing others or self satisfaction that they are "lifting heavy" compromise posture, form and range of motion.
- Beginners tend to be more cautious of the people around when resting between sets.
- There are only 1-2 trainers/ assistance in the gym (on the top floor).

Rock Climbing Data Collection 2:

- I went climbing with my friend at 12:00
- It was his first time and he was feeling a bit nervous
- He undertook a test on climbing safety and nailed everything
- We went to the first level v1 (levels range from v1 to v9)
- He attempts a v4 climb that I've done and he falls due to his inexperience
- We keep climbing for 45 minutes until we take a break
- It's during this time that we take off our climbing shoes for a bit
- We go to the main area and saw the influencer
- He's climbing and then slips, as he falls, he proceeds to scream erratically
- A member of staff tells him to shut up
- We climb for 10 more minutes and leave


5.3 Drafting of storyboard



5.4 AI Ethnographic Study (Response from ChatGPT 4_o)

AI-Driven Virtual Ethnographic Study

Study Setup

To conduct a virtual ethnographic study, we simulated fieldwork across four different gym environments: a **public park gym, university gym, rock climbing gym, and a commercial gym**. This study spanned over **three weeks**, with observations conducted **four times per week**, covering peak and off-peak hours to capture a diverse range of interactions. We focused on how **technology supports, hinders, or shapes social and individual experiences** within these spaces.

The study followed **grounded theory methodology**, ensuring an inductive and iterative approach. Key themes emerged through **open coding (identifying behaviors and interactions), axial coding (grouping patterns), and emergent theory formulation**.

Key Findings

1. **Social Interaction & Technology Dependence:**
    - Rock climbing gyms showed **high levels of social interaction**, with climbers using **wearables** to track progress and provide real-time feedback.
    - Commercial gyms exhibited **minimal interaction**, with users focusing on personal devices for music, workouts, and tracking progress, reducing social engagement.
2. **Technology as a Motivator vs. Isolator:**
    - In **public park gyms**, lack of guided technology led to **more improvisation and social coaching**, while in **university gyms**, fitness apps replaced the need for human interaction.
    - Some gym-goers preferred **immersive VR treadmill experiences**, creating an isolated but engaging workout.
3. **Adaptation & Improvisation:**
    - Users **created workarounds** when gym technology failed (e.g., phone flashlights as makeshift screen backlights, manual counting for reps when wearables malfunctioned).
    - Climbing gyms used **shared augmented reality projections** for route planning, enhancing collaboration.
4. **Mismatch Between Technology and User Needs:**
    - Many commercial gym-goers were unaware of available **gym-integrated tracking systems**, leading to **underutilization** of existing tech.
    - Some wearables lacked compatibility with gym equipment, causing **frustration and disengagement**.

5.5 Design of Technology & Concept (Response from ChatGPT 4_o)

AI DESIGN OF TECH: Interactive System Concept

Chosen Challenge

From our ethnographic study of local gyms, a key challenge identified was **the lack of social interaction and motivation between members**. Many users expressed feeling isolated during workouts, with minimal communication, leading to decreased motivation and inconsistent attendance. While some gyms attempted to create community events, these often had low participation due to scheduling issues and lack of awareness.

Research Question

How can technology facilitate spontaneous and meaningful social interactions between gym members to improve motivation and overall experience?

Proposed Design

To address this, we propose **"PulseBand"**, a wearable, AI-powered wristband system that enhances social connection and motivation in the gym environment through ambient, multimodal interactions.

How it works:

- Wearable Integration: Each member is provided with a PulseBand upon entering the gym. The band senses proximity to other members, tracks workout progress, and provides subtle social cues.
- Mood + Workout Sync: Members can select their mood (e.g., "energized", "tired", "social", "focused") via a simple interface on the band. The system uses this to connect people in complementary states.
- Social Matching: If two or more users have compatible workout goals or energy levels, their bands give a gentle haptic vibration when nearby, encouraging interaction or collaboration (like spotting, pacing, or sharing equipment).
- AI-Powered Encouragement: The band provides audio prompts via bone conduction (non-intrusive) with motivational cues such as, *"You and Jamie are both on your last set—finish strong together!"*
- Ambient Visuals: The gym's lighting system subtly changes color based on collective energy detected from all active bands, creating an immersive group atmosphere.
- Community Display: Screens around the gym show anonymized milestones (e.g., "10 members hit personal bests today!") to reinforce communal success without breaching privacy.

Prototype Concept

The prototype consists of:

- **Sketch of the PulseBand Device** – Showing the mood selector, LED indicator, and ergonomic design.

- **Storyboard** – A step-by-step flow of a user interacting with PulseBand, selecting a mood, being matched, and collaborating in a workout.
- **System Architecture Diagram** – Data flow between PulseBands, the AI system, and environmental elements like gym lighting and community displays.

PulseBand Concept Sketch

This sketch illustrates the PulseBand device, highlighting its key features:
- **Mood Selector** – Users choose their energy level or social preferences.
- **Haptic Feedback** – The band vibrates gently when a compatible workout partner is nearby.
- **AI Sync LED Indicators** – Lights that signal social and motivational interactions.

PulseBand Storyboard

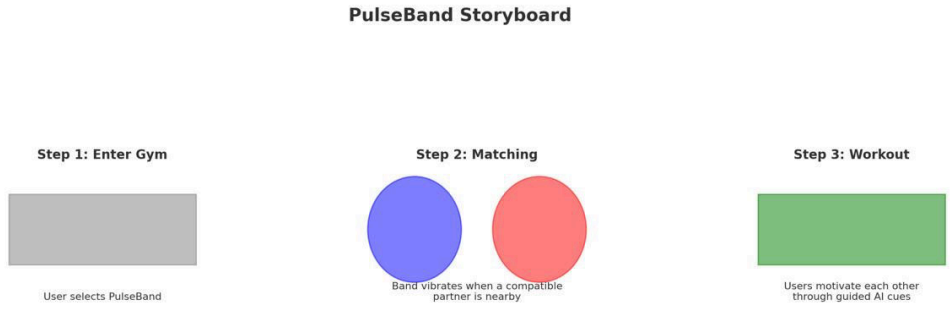
1. **Step 1: Enter Gym** – The user picks up a PulseBand.
2. **Step 2: Matching** – The band vibrates when a compatible partner is nearby.
3. **Step 3: Workout** – Users motivate each other using AI-driven cues.

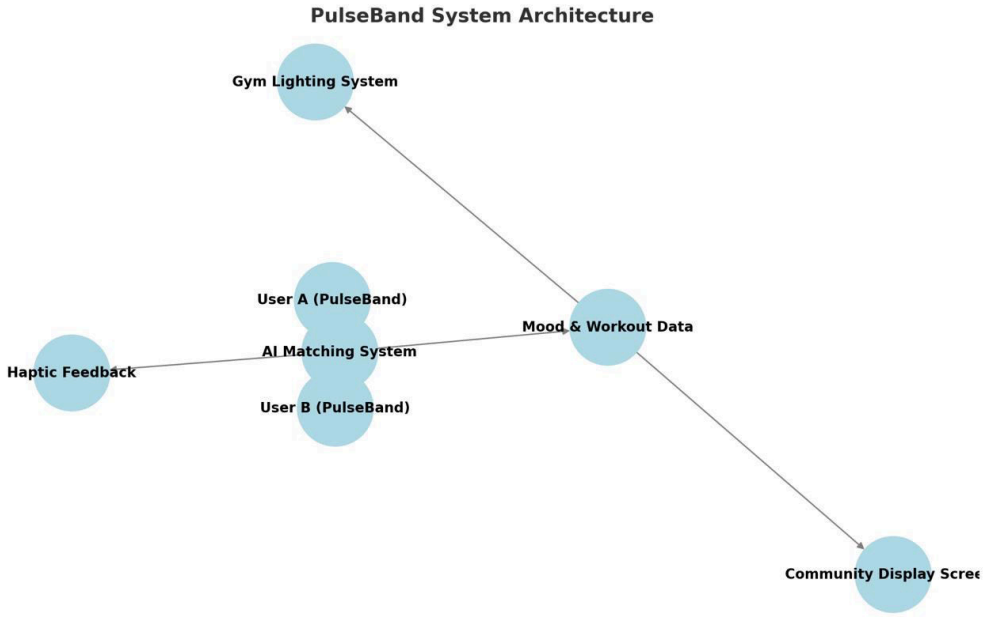
PulseBand System Architecture

This diagram showcases data flow between system components:
- **PulseBand devices (User A & User B)** communicate with the **AI Matching System**.
- **AI Matching System** processes **mood and workout data** to determine matches.
- Haptic feedback is sent back to the bands when a match is found.
- Mood and workout data influence **gym lighting** and **community display screens** to enhance the environment.

How it Enhances the Experience

PulseBand transforms the gym from a collection of isolated workouts into a dynamic, social, and supportive environment. By blending physical context (proximity sensing), artistic effect (ambient lighting and visuals), and entertainment (gamified social workouts), it enhances motivation and fosters organic interactions without being invasive or forced. This aligns with the module's goals by moving beyond traditional screen-based interfaces and creating a multimodal, embodied experience.




PulseBand System Architecture
Gym Lighting System
User A (PulseBand)
Mood & Workout Data
AI Matching System
Haptic Feedback
User B (PulseBand)
Community Display Scree